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(b) Identify the processes in which change in internal energy is least and is maximum.
(c) Show each process on a PV diagram.
(d) Name the processes in which the heat transfer is maximum and minimum.

(Take $\gamma = \frac{5}{3}$ and R=8.3 J mol^{-1} K^{-1})

Solution

(a) (i) In an adiabatic process the work done by the system is

$$W_{adia} = \frac{\mu R}{\gamma - 1}[T_i - T_f]$$

To find the final temperature T_f, we can use adiabatic equation of state.

$$T_f V_f^{\gamma-1} = T_i V_i^{\gamma-1}$$

$$T_f = T_i \left(\frac{V_i}{V_f}\right)^{\gamma-1} = 300 \times \left(\frac{1}{2}\right)^{\frac{2}{3}}$$

$$= 0.63 \times 300K = 189.8K$$

$$W = 1 \times 8.3 \times \frac{3}{2}(300 - 189.8) = 1.37kJ$$

(ii) In an isobaric process the work done by the system

$$W = P\Delta V = P(V_f - V_i)$$

and $V_f = 2V_i$ so $W = 2PV_i$

To find V_i, we can use the ideal gas law for initial state. $P_i V_i = RT_i$

$$V_i = \frac{RT_i}{P_i} = 8.3 \times \frac{300}{1} \times 10^{-6} = 24.9 \times 10^{-4} m^3$$

The work done during isobaric process,
$W = 2 \times 10^6 \times 24.9 \times 10^{-4} = 4.9\ kJ$

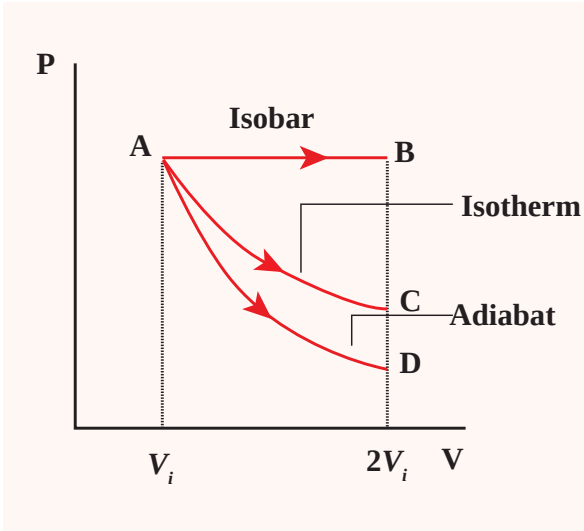
(iii) In an isothermal process the work done by the system,

$$W = \mu RT\ ln\left(\frac{V_f}{V_i}\right)$$

In an isothermal process the initial room temperature is constant.

$$W = 1 \times 8.3 \times 300 \times ln(2) = 1.7kJ$$

(b) Comparing all three processes, we see that the work done in the isobaric process is the greatest, and work done in the adiabatic process is the least.
(c) The PV diagram is shown in the Figure.



The area under the curve AB = Work done during the isobaric process
The area under the curve AC = Work done during the isothermal process
The area under the curve AD= Work done during the adiabatic process
From the PV diagram the area under the curve AB is more, implying that the work done in isobaric process is highest and work done in adiabatic process is least.
(d) In an adiabatic process no heat enters into the system or leaves from the system. In an isobaric process the work done is more so heat supplied should be more compared to an isothermal process.

8.8.4 Isochoric process

This is a thermodynamic process in which the volume of the system is kept constant. But pressure, temperature and internal energy continue to be variables.